

Below are the detailed descriptions for the algorithmic trading system, including trading strategies, position sizing, money management rules, technical indicators, market watch, frontend, backend, AI Agents, etc. to develop the app by using the Interactive Brokers Paper Trading API. The app is to be developed in Python, leveraging the Polars library, DuckDB, and if required, any other libraries, databases, or frameworks, for efficient data manipulation and analysis. The app will execute paper / simulated trades across multiple asset classes, including stocks, indices, forex, and commodities from exchanges like NYSE, NASDAQ, LSE, and NSE.

1. Trend Following Strategy:

Detailed Trading Strategy Description:

Strategy Overview

This is a positional, trend-following trading strategy designed to capture long-term price movements in various asset classes, including stocks, indices, forex, and commodities. It uses volume-weighted technical indicators to identify trends, momentum, and volatility, ensuring trades align with market strength while managing risk effectively. The strategy below is for long-only and I would like you to adapt it for short positions. It should be implemented in Python using the Polars library for data processing and the Interactive Brokers Paper Trading API for simulated execution.

- Type: Positional, trend-following strategy (long-only).
- Assets: Stocks, indices, forex, commodities.
- Libraries: Polars for data processing, DuckDB, is required, Interactive Brokers API for trading.

Key Technical Indicators

The strategy relies on the following volume-weighted indicators, calculated using historical price and volume data:

- 55-day Volume-Weighted Simple Moving Average (VW SMA) of Open:
 - Identifies the long-term trend direction.
 - Volume-weighted moving average of opening prices over 55 days.
- 13-day VW SMA of Open:
 - Tracks medium-term price movements (optional for additional confirmation).
 - Volume-weighted moving average of opening prices over 13 days (optional).
- 14-day Volume-Weighted Money Flow Index (MFI) of HLC Average:
 - Measures momentum based on the average of High, Low, and Close prices.
 - Volume-weighted Money Flow Index based on the average of High, Low, Close prices over 14 days.
- 34-day VW SMA of MFI(14):
 - Smooths the MFI for trend confirmation.

- Smoothing of the 14-day MFI over 34 days.
- 21-day VW SMA of MFI(14):
 - Used as an exit trigger.
 - Smoothing of the 14-day MFI over 21 days.
- 5-day VW SMA of High:
 - Defines short-term resistance levels.
 - Volume-weighted moving average of high prices over 5 days.
- 5-day VW SMA of Low:
 - Defines short-term support levels for exits.
 - Volume-weighted moving average of low prices over 5 days.
- 21-day Volume-Weighted Average True Range (ATR):
 - Measures volatility to set stop-loss and take-profit levels.
 - Volume-weighted Average True Range over 21 days.

Let me know if you need the formula for the Volume Weighted indicators.

Entry Rules

A long position is initiated when both of the following conditions are met:

- The 14-day MFI is greater than the 34-day VW SMA of MFI, indicating strong momentum.
- The current price is greater than the 55-day VW SMA of Open, confirming an uptrend.
- Long Position:
 - 14-day VW MFI > 34-day VW SMA of MFI(14).
 - Current price > 55-day VW SMA of Open.
- Trigger a buy order via the Interactive Brokers API when conditions are met.

A short position is initiated when both of the following conditions are met:

- The 14-day MFI is lower than the 34-day VW SMA of MFI, indicating strong momentum.
- The current price is lower than the 55-day VW SMA of Open, confirming a downtrend.
- Short Position:
 - 14-day VW MFI < 34-day VW SMA of MFI(14).
 - Current price < 55-day VW SMA of Open.
- Trigger a sell order via the Interactive Brokers API when conditions are met.

Note: If not specified, the short positions should use inverse conditions of the long positions. For the sake of convenience and to avoid writing the inverse for the short position, the strategy below may focus only on long trades. The inverse of the long strategy will have to be done to take the short position.

Exit Rules

Long Position:

Positions are closed based on the following criteria:

- Stop-Loss: Set at 2x the 21-day VW ATR below the entry price to limit downside risk.

(Stop-Loss: $2 \times 21\text{-day VW ATR}$ below entry price)

- Take-Profit: Set at 2x the 21-day VW ATR above the entry price to secure profits.

(Take-Profit: $2 \times 21\text{-day VW ATR}$ above entry price)

- Additional Exit Conditions:

- Exit if the 14-day MFI falls below the 21-day VW SMA of MFI, signalling weakening momentum.

(14-day VW MFI < 21-day VW SMA of MFI(14))

- Exit if the current price drops below the 5-day VW SMA of Low, indicating a breakdown of short-term support.

(Current price < 5-day VW SMA of Low)

Short Position:

Positions are closed based on the following criteria:

- Stop-Loss: Set at 2x the 21-day VW ATR above the entry price to limit downside risk.

(Stop-Loss: $2 \times 21\text{-day VW ATR}$ above entry price)

- Take-Profit: Set at 2x the 21-day VW ATR below the entry price to secure profits.

(Take-Profit: $2 \times 21\text{-day VW ATR}$ below entry price)

- Additional Exit Conditions:

- Exit if the 14-day MFI rises above the 21-day VW SMA of MFI, signalling weakening momentum.

(14-day VW MFI > 21-day VW SMA of MFI(14))

- Exit if the current price rises above the 5-day VW SMA of Low, indicating a breakdown of short-term support.

(Current price > 5-day VW SMA of Low)

Note:

- Place stop-loss and take-profit orders with the API upon entry; monitor additional exit conditions in real time.

Initial Account Balance: £10,000/-

Position Sizing

Position sizing is calculated to limit risk to 1% of the total account balance per trade, adjusted for the asset's volatility and lot size.

(Risk per Trade: 1% of account balance)

The formula is:

$$\text{Position Size} = \frac{\text{Account Balance} \times 0.01}{2 \times N \times \text{Lot Size}}$$

which is:

$$\text{Position Size} = \frac{\text{Account Balance} \times 0.01}{2 \times \text{21-day VW ATR} \times \text{Lot Size}}$$

Where:

- Account Balance: Total capital in the trading account (e.g., £10,000).
 - N: The 21-day VW ATR, representing per-unit risk.
 - Lot Size: Varies by asset (Adjust based on asset):
 - Stocks: 1 share.
 - Forex: 100,000 units of the base currency (standard lot).
 - Commodities/Futures: Depends on the contract (e.g., 100 oz for gold, 1,000 barrels for oil).
 - Indices: Depends on the contract multiplier (e.g., 10 for S&P 500 futures).
- (Commodities/Indices: Per contract specification (e.g., 100 oz for gold))
- Calculate dynamically using account balance and real-time ATR.

Example:

- Account Balance = £10,000.
- Risk = 1% = £100.
- Asset = Stock, Lot Size = 1 share.
- Entry Price = £50, 21-day VW ATR = £2.
- Stop-Loss Distance = $2 \times £2 = £4$.
- Position Size = $£100 / (2 \times £2 \times 1) = 25$ shares.

Money Management Rules

The strategy incorporates the following rules to protect capital and optimise returns:

- Maximum Drawdown Limit: If the account experiences a drawdown exceeding 20%, reduce the risk per trade to 0.5% (i.e., halve the position size formula's risk factor).

(Drawdown Limit: If account drawdown > 20%, adjust risk to 0.5% per trade)

- Trailing Stop: Implement a trailing stop set at 1.5x the 21-day VW ATR below the highest price reached since entry. This locks in profits as the price rises while allowing room for normal volatility.

(Trailing Stop: Set at $1.5 \times 21\text{-day VW ATR}$ below the highest price since entry; update via the API as price moves)

- Currency Adjustments: Convert all risk calculations to the account's base currency (e.g., USD for US markets, GBP for LSE) to ensure consistency across assets.

(Currency Handling: Convert all risk calculations to the account's base currency (e.g., USD, GBP, etc.))

Assets Traded

The strategy applies to a diversified portfolio:

- Stocks: Major companies from NYSE, NASDAQ, LSE, and NSE.
- Indices: Key indices like S&P 500, NASDAQ 100, DJI, FTSE 100, or Nifty 50.
- Forex: Major pairs (e.g., EUR/USD, USD/JPY).
- Commodities: Gold, Crude Oil (WTI), Silver.

API Integration

- Data Feed: Fetch real-time OHLCV (Open, High, Low, Close, Volume) data from Interactive Brokers Paper Trading API.
- Order Execution: Place market orders for entries and limit/stop orders for exits.
- Simulation: Ensure all trades are executed in paper trading mode.

Implementation Notes

- Use Polars to compute indicators efficiently on large datasets.
- Use DuckDB, if required.
- Handle multiple assets concurrently, adjusting lot sizes and currency conversions.
- Include error handling for API connectivity and data issues.
- Log all trades and performance metrics (e.g., profit/loss, drawdown) for analysis.

Deliverables

Generate Python code that:

1. Connects to the Interactive Brokers Paper Trading API.
2. Calculates all specified indicators using Polars.
3. Implements the entry, exit, position sizing, and money management rules.

4. Executes and monitors trades in real time.
5. Supports the listed asset classes with appropriate adjustments.

Please ensure the code is modular, well-commented, and compatible with VS Code debugging.

After the code is written, create Unit Testing to test the strategy in detail.

Additional Notes

- If you feel that something else needs to be added (e.g., adjusting indicator periods, or targeting specific assets) to refine the strategy, please let me know.
- Let me know if you require any specific inputs, information or clarification.

Since this trading strategy is a detailed one, I would like you to let me know if this should be included as part of the Dashboard and the Market Watch application that we started with OR if this should be made into another application?

2. Support and Resistance Strategy Using Gann Fan, Fibonacci, and Pivot Points

Trading Strategy:

Here is a comprehensive support and resistance trading strategy that incorporates Gann Fan, Fibonacci retracement, and Pivot Points for both long and short positions. This strategy includes detailed rules for entry (buy/sell), exit (take profit/stop loss), position sizing, money management, and trailing stop loss. It is designed for positional trading and can be applied to various assets (e.g., stocks, forex, commodities) on daily or weekly charts.

1. Daily Preparation: Identifying Key Levels

Before trading, calculate and plot the following technical tools:

- Pivot Points:
 - Use the previous period's high, low, and close to determine the Pivot Point (PP), support levels (S1, S2), and resistance levels (R1, R2).
 - Formulas:
 - $$\text{PP} = \frac{\text{High} + \text{Low} + \text{Close}}{3}$$
 - $$\text{R1} = 2 \times \text{PP} - \text{Low}$$
 - $$\text{S1} = 2 \times \text{PP} - \text{High}$$
 - $$\text{R2} = \text{PP} + (\text{High} - \text{Low})$$
 - $$\text{S2} = \text{PP} - (\text{High} - \text{Low})$$
- Fibonacci Retracement Levels:
 - Identify a recent significant price swing (e.g., from a swing low to a swing high).
 - Plot Fibonacci levels at 23.6%, 38.2%, 50%, 61.8%, and 100% to identify potential support and resistance zones.
- Gann Fan:
 - Draw a Gann Fan from a significant high or low point.
 - Key angles (e.g., 1x1 (45°), 1x2, 2x1) act as dynamic support and resistance lines based on price-time relationships.

These tools together create a map of key price levels to monitor for trading opportunities.

2. Entry Rules

Long Position (Buy)

- Condition 1: Price breaks above a resistance level, such as R1 (from Pivot Points) or a key Fibonacci level (e.g., 61.8%).
- Condition 2: Price finds support at a Gann Fan angle (e.g., bounces off the 1x1 angle) during an uptrend.
- Optional Confirmation: Look for bullish signals like a candlestick pattern (e.g., hammer, bullish engulfing) or rising volume.

Short Position (Sell)

- Condition 1: Price breaks below a support level, such as S1 (from Pivot Points) or a key Fibonacci level (e.g., 38.2%).
- Condition 2: Price meets resistance at a Gann Fan angle (e.g., rejected at the 1x1 angle) during a downtrend.
- Optional Confirmation: Look for bearish signals like a candlestick pattern (e.g., shooting star, bearish engulfing) or rising volume.

3. Exit Rules

Take Profit

- Long Position:
 - Set at the next resistance level, such as R2 (Pivot Points), the next Fibonacci level (e.g., 100%), or the next Gann Fan angle (e.g., 2x1).
- Short Position:
 - Set at the next support level, such as S2 (Pivot Points), the next Fibonacci level (e.g., 0%), or the next Gann Fan angle (e.g., 1x2).

Stop Loss

- Long Position:
 - Place below the recent support level, such as S1 or the most recent swing low.
 - Alternatively, use a fixed percentage (e.g., 1-2% below entry price).
- Short Position:
 - Place above the recent resistance level, such as R1 or the most recent swing high.
 - Alternatively, use a fixed percentage (e.g., 1-2% above entry price).

4. Position Sizing

- Risk per Trade: Risk a fixed percentage of your account balance, typically 1-2%.
- Formula:
$$\text{Position Size} = \frac{\text{Account Balance} \times \text{Risk \%}}{|\text{Entry Price} - \text{Stop Loss Price}|}$$
 - This determines the number of units (e.g., shares, lots) to trade based on the distance between entry and stop loss.
- Adjust for Asset: Round to the nearest whole unit or meet the asset's minimum trade size requirements.

Example:

- Account Balance: \$10,000
- Risk: 1% (\$100)
- Entry: \$52.10
- Stop Loss: \$47.90 (distance = \$4.20)
- Position Size: $\left(\frac{100}{4.20} \right) \approx 23.8 \rightarrow$ Trade 23 shares.

5. Money Management

- Maximum Risk per Trade: Limit risk to 1-2% of your account balance per trade.
- Total Portfolio Risk: Keep total open risk across all trades below 5-10% of your account balance.
- Diversification: Avoid concentrating risk in a single asset or market; trade across different sectors or instruments.
- Volatility Adjustment: Use the Average True Range (ATR) to scale position sizes—smaller positions for high-volatility assets, larger for low-volatility ones.

6. Trailing Stop Loss

- Percentage-Based:
 - Long Position: Set at 2-3% below the highest price reached since entry.
 - Short Position: Set at 2-3% above the lowest price reached since entry.
- Volatility-Based:
 - Use 2x ATR (e.g., 14-period ATR) below the current price for longs or above for shorts.
- Purpose: Locks in profits as the price moves favorably while allowing room for normal price fluctuations.

Example (Long):

- Entry: \$52.10
- Price rises to \$55 → Trailing Stop: $\$55 - 2\% = \53.90 .
- Adjust upward as the price continues to rise.

Example Scenarios

Long Trade Example

- Asset: Stock XYZ
- Account Balance: \$10,000
- Pivot Points: PP = \$50, R1 = \$52, S1 = \$48
- Fibonacci: 38.2% at \$51
- Gann Fan: 1x1 angle at \$50.50
- Entry: Price breaks R1 (\$52) and holds above the Gann 1x1 angle at \$52.10.
- Stop Loss: Below S1 at \$47.90 (distance = \$4.20).
- Take Profit: At R2 (\$54) or next Fibonacci level.
- Position Size: $\left(\frac{100}{4.20}\right) \approx 23$ shares.
- Trailing Stop: Adjust to 2% below the highest price (e.g., \$53.90 if price hits \$55).

Short Trade Example

- Asset: Stock XYZ
- Account Balance: \$10,000
- Pivot Points: PP = \$50, R1 = \$52, S1 = \$48
- Fibonacci: 61.8% at \$49
- Gann Fan: 1x1 angle at \$50.50
- Entry: Price breaks below S1 (\$48) and is rejected at the Gann 1x1 angle at \$47.90.
- Stop Loss: Above R1 at \$52.10 (distance = \$4.20).
- Take Profit: At S2 (\$46) or next Fibonacci level.
- Position Size: $\left(\frac{100}{4.20}\right) \approx 23$ shares.
- Trailing Stop: Adjust to 2% above the lowest price.

Additional Considerations

- Timeframes: Use daily charts for shorter-term trades. Adjust Pivot Point calculations accordingly.
- Confirmation Tools: Add indicators like VW MFI, VW RSI, VW MACD, etc. or volume for stronger entry/exit signals.

- Market Context: Avoid trading during major news events or extreme volatility, as support/resistance may break unexpectedly.
- Backtesting: Test the strategy on historical data to optimize parameters like stop-loss distance or take-profit targets.

Implementation Checklist

1. Plot Levels: Calculate and mark Pivot Points, Fibonacci retracements, and Gann Fan angles daily.
2. Monitor Price: Watch for breakouts or reversals at these levels to enter trades.
3. Set Orders: Place entry, stop-loss, and take-profit orders based on the rules.
4. Manage Risk: Calculate position size and ensure total risk stays within limits.
5. Trail Profits: Adjust the trailing stop as the trade moves in your favor.
6. Review: Analyse trade outcomes to refine the strategy over time.

This strategy provides a structured, risk-managed approach to trading support and resistance using Gann Fan, Fibonacci, and Pivot Points. It balances technical precision with practical money management, making it suitable for both long and short positions.